ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** several branches of science together	real-world problems cross disciplinary boundaries
2	**(b)** use relationships and check with an estimate	reasoning first, calculation second
3	**(d)** large enough to be seen and to settle	particles settle on standing
4	**(d)** a beam of light passing through fog	colloidal particles scatter light
5	**(a)** phosphorus trichloride	prefixes show the number of atoms; second element ends in -ide
6	**(c)** ionic compounds	no discrete molecules in ionic solids
7	**(d)** 32	$2n^2 = 2 \times 4^2 = 32$
8	**(b)** 18 m s^{-1}	900 ÷ 50
9	**(d)** acceleration	rate of change of velocity
10	**(d)** 10 kg m s^{-1}	p = mv
11	**(a)** 4 m s^{-2}	a = F/m
12	**(d)** 480 J	W = F × s
13	**(b)** 20 Hz	oscillations ÷ time
14	**(b)** lysosomes	breaks down worn-out parts
15	**(a)** cellulose	carbohydrate
16	**(b)** ligaments	elastic tissue at joints
17	**(a)** Echinodermata	spiny skin
18	**(b)** sperms	produced in the testes
19	**(b)** cutting	artificial vegetative propagation
20	**(c)** 40 kWh	energy = power × area × time

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Approximate: estimating the rice needed for a month. Exact: the dose of a medicine (1+1).	1+1
22	Na (2,8,1) loses one electron to form Na^+; Cl (2,8,7) gains it to form Cl^-; the ions attract (2).	1+1
23	The direction of velocity changes continuously, so velocity changes (2).	2
24	Friction is very small, so the foot cannot push the ground backwards and the ground cannot push forward enough (2).	2
25	They break down worn-out organelles and foreign material using enzymes (2).	2
26	Protection, secretion, absorption and sensory functions (2).	2
27	External: outside the body in water; internal: inside the female body (1+1).	1+1
28	Rhizobium bacteria and lightning fix N_2 into nitrogen compounds absorbed by plants (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	The solubility of gas (CO_2) in liquids decreases with temperature (2), so the gas escapes (1).	2+1
30	Protium (1p, 0n), deuterium (1p, 1n), tritium (1p, 2n) (1+1+1).	1+1+1
31	Scissors: fulcrum between (1); wheelbarrow: load between (1); tweezers: effort between (1).	1+1+1
32	In a small room reflections return in less than 0.1 s so are not separated (1.5); in a hall the gap is longer, so echoes are heard (1.5).	1½+1½
33	They contract rhythmically without fatigue (1.5); they are branched and involuntary, while skeletal muscles are unbranched and voluntary (1.5).	1½+1½
34	Flatworms: flat, bilateral, no body cavity; roundworms: cylindrical, with a body cavity and complete gut (3).	3

35	Mg (2,8,2) loses two electrons to form Mg^{2+} (1); each Cl gains one electron: two Cl^{-} (1); formula $MgCl_2$ — two chloride ions per magnesium ion (1).	1+1+1
	OR: NaCl forms free Na^{+} and Cl^{-} ions in water which carry current (2); sugar is covalent and does not produce ions (1).	2+1
36	No food or oxygen production; the CO_2 would rise and food chains would collapse (3).	3
	OR: Enhanced greenhouse effect (1); global warming, glacier melting, sea-level rise, extreme weather (any two) (2).	1+2

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) A compound always contains the same elements in a fixed mass ratio; Proust (1). (ii) Water is a compound; its composition does not depend on the source (1). (iii) H : O = 1 : 8 → H 4 g, O 32 g (2). (iv) No — mixtures can have any ratio (1).	1+1+2+1
	OR: (i) A bond formed by sharing electron pairs (1). (ii) Each H shares 1 e^{-} forming H—H; each Cl shares 1 e^{-} forming Cl—Cl; H and Cl share 1 pair forming H—Cl (3). (iii) O_2: two pairs (double bond); N_2: three pairs (triple bond) (1).	1+3+1
38	(i) Net force is zero, the block stays at rest (1). (ii) 5 N (1). (iii) 0.2 m s^{-2} (1). (iv) ½ × 0.2 × 4 = 0.4 m (2).	1+1+1+2
	OR: (i) A cart pulled by a thread over a pulley with cup masses: doubling the load in the cup increases acceleration (2). (ii) Heavier cart, smaller acceleration (1). (iii) a = F/m or F = ma (2).	2+1+2
39	(i) Controls cell activities and stores DNA (1). (ii) Chromatin is entangled DNA threads; condenses into chromosomes; genes are functional segments of DNA (3). (iii) They lack a nucleus (1).	1+3+1
	OR: (i) All living things consist of cells; the cell is the unit of structure and function; cells arise from cells (2). (ii) Schleiden, Schwann and Virchow (1). (iii) All life processes occur in cells (2).	2+1+2
40	(i) Self and cross (1). (ii) Wind, water, insects, birds with flower features (2). (iii) Pollen tube grows, fertilisation, seed and fruit form (2).	1+2+2
	OR: (i) Bread mould (Rhizopus) (1). (ii) Spores in the air (1). (iii) By spore formation (1). (iv) Penicillin antibiotic; decomposing wastes (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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